KAIZEN MATHS

Lesson starter

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name

S1

Differentiate with respect to x.

$f(x) = 4x^2 + 6x - 2$

S2

Differentiate with respect to x.

$f(x) = 4x^3 + 8x^2 - 9x + 11$

S3

Differentiate with respect to x.

$f(x) = 3x^5 + 4x^4 - 5x^2 + 7$

S4

Differentiate with respect to x.

$f(x) = 5x^2 - 2x - 15$

Show your working

KAIZEN MATHS

Lesson starter

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name

S1

Differentiate with respect to x.

$f(x) = 4x^2 + 6x - 2$

S2

Differentiate with respect to x.

$f(x) = 4x^3 + 8x^2 - 9x + 11$

S3

Differentiate with respect to x.

$f(x) = 3x^5 + 4x^4 - 5x^2 + 7$

S4

Differentiate with respect to x.

$f(x) = 5x^2 - 2x - 15$

ALG-E739-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name _____

E1

Differentiate with respect to x.

$f(x) = 5x^3 + 8x^2 - 2x + 2$

E2

Differentiate with respect to x.

$f(x) = 4x^4 + 2x^3 - 2x^2 + 8$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 11x - 2$

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name _____

E1

Differentiate with respect to x.

$f(x) = 5x^3 + 8x^2 - 2x + 2$

E2

Differentiate with respect to x.

$f(x) = 4x^4 + 2x^3 - 2x^2 + 8$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 11x - 2$

ALG-E739-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name _____

E1

Differentiate with respect to x.

$f(x) = 5x^3 + 8x^2 - 2x + 2$

E2

Differentiate with respect to x.

$f(x) = 4x^4 + 2x^3 - 2x^2 + 8$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 11x - 2$

ALG-E739-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Multi-Term Polynomials

Name _____

E1

Differentiate with respect to x.

$f(x) = 5x^3 + 8x^2 - 2x + 2$

E2

Differentiate with respect to x.

$f(x) = 4x^4 + 2x^3 - 2x^2 + 8$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 11x - 2$

ALG-E739-L2 · E

Show your working

Teacher answers & working

Basic Differentiation of Polynomials · Multi-Term Polynomials

ALG-E739-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Differentiate with respect to x.

$f(x) = 4x^2 + 6x - 2$

Answer:

$f'(x) = 8x + 6$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

S2

Differentiate with respect to x.

$f(x) = 4x^3 + 8x^2 - 9x + 11$

Answer:

$f'(x) = 12x^2 + 16x - 9$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Teacher answers & working

Basic Differentiation of Polynomials · Multi-Term Polynomials

ALG-E739-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Differentiate with respect to x.

$f(x) = 3x^5 + 4x^4 - 5x^2 + 7$

Answer:

$f'(x) = 15x^4 + 16x^3 - 10x$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

S4

Differentiate with respect to x.

$f(x) = 5x^2 - 2x - 15$

Answer:

$f'(x) = 10x - 2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Teacher answers & working

Basic Differentiation of Polynomials · Multi-Term Polynomials

ALG-E739-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Differentiate with respect to x.

$f(x) = 5x^3 + 8x^2 - 2x + 2$

Answer:

$f'(x) = 15x^2 + 16x - 2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Teacher answers & working

Basic Differentiation of Polynomials · Multi-Term Polynomials

ALG-E739-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E2

Differentiate with respect to x.

$f(x) = 4x^4 + 2x^3 - 2x^2 + 8$

Answer:

$f'(x) = 16x^3 + 6x^2 - 4x$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 11x - 2$

Answer:

$f'(x) = 8x + 11$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.